# Aerospace Program Portfolio: Findings and Recommendations

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## Bottom line

The six-program portfolio ran **$2.62M (1.24%) over budget** from January 2024 to December 2025, and **two programs caused 93% of it**.

- **Sentinel** is over budget in Labor since August 2024, and the monthly overrun grew from about $30K in late 2024 to about $70K in late 2025. It is the most urgent problem.
- **Orion** is the largest overrun, $1.56M, driven by Materials since June 2024.
- **Atlas** looks on budget, but it has the weakest cost efficiency in the portfolio. It is an early warning, not yet an overrun.

Prepared for the VP Finance and Program Finance team. This is a simulated scenario built on synthetic data (see Limitations).

## 1. Portfolio performance: $2.62M over budget, status Monitor

Against a $212.19M budget, the portfolio spent $214.81M, a net overrun of $2,621,451 (1.24%). That puts the portfolio in the **Monitor** band (1.0% to 2.5% over budget).

Earned value tells the same story. The portfolio's CPI is 0.979: it delivered about 98 cents of planned work for every dollar spent, a $4.41M cost variance.

The $2.62M is a **net** figure. Four programs ran over by $2.74M combined, and Falcon and Atlas offset $117K of that by coming in under budget.

## 2. Program drivers: Orion and Sentinel cause 93% of the overrun

Orion and Sentinel together account for $2.45M of the $2.62M net overrun. The other four programs are within 0.6% of budget.

| Program | Over / (under) budget | Variance % | Share of net overrun | CPI | Management Status | CPI Watch |
|---|---|---|---|---|---|---|
| Orion | $1,557,692 | 3.49% | 59.4% | 0.964 | High Attention | Watch |

| Program | Over / (under) budget | Variance % | Share of net overrun | CPI | Management Status | CPI Watch |
|---|---|---|---|---|---|---|
| Sentinel | $887,639 | 2.56% | 33.9% | 0.974 | High Attention | Watch |
| Horizon | $190,544 | 0.52% | 7.3% | 0.993 | On Track | On Track |
| Vanguard | $102,836 | 0.40% | 3.9% | 0.991 | On Track | On Track |
| Falcon | ($45,210) | (0.11%) | (1.7%) | 1.001 | On Track | On Track |
| Atlas | ($72,050) | (0.24%) | (2.7%) | 0.954 | On Track | Watch |
| **Portfolio** | **$2,621,451** | **1.24%** | **100%** | **0.979** | **Monitor** | **Monitor** |

Parentheses mean under budget. Shares add to 100% only with the negatives included, because the total is a net figure.

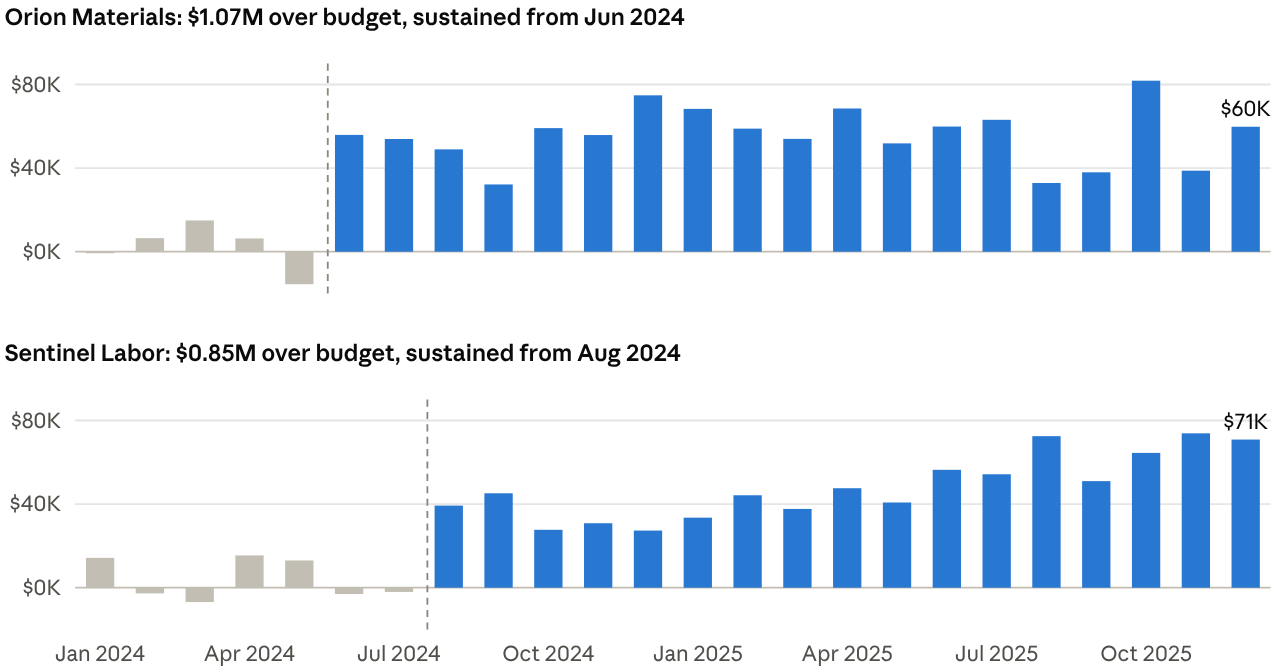
## 3–4. Cost drivers and timing: one category per program, both since mid-2024

Each overrun traces to a single cost category:

- **Orion: Materials**, $1,068,740, or 68.6% of Orion's overrun. Subcontracts adds another $373,775 (24.0%).
- **Sentinel: Labor**, $845,459, or 95.2% of Sentinel's overrun. Its other five categories add only $42K combined.

**Both driver overruns became sustained in mid-2024, and Sentinel Labor is still growing**

Monthly over / (under) budget of the main cost driver in each program. Blue = months of sustained overrun.




Power BI model, validated against SQL · Orion Materials and Sentinel Labor, Jan 2024 to Dec 2025

The two overruns behave differently. Orion's Materials overrun has run at $32K to $82K a month since June 2024 with no clear trend. Sentinel's Labor overrun has climbed steadily, from about $30K a month in late 2024 to $64K to $74K a month in October to December 2025.

## 5. Management attention: two programs now, one early warning

Orion and Sentinel are **High Attention**, at 3.49% and 2.56% over budget. Both also fall below the CPI Watch line of 0.975, so the budget view and the earned-value view agree on them.

**Atlas is the exception the budget view misses.** It spent $29.66M against a $29.74M budget, so it looks $72K under budget. But it earned only $28.30M of planned work, a $1.37M cost variance that is the worst in the portfolio and larger than Sentinel's ($0.92M). Its CPI of 0.954 means it delivers about 95 cents of planned work per dollar spent.

In plain terms, Atlas is spending on plan but falling behind on work. If its efficiency stays at this level, finishing the planned work will cost more than budgeted, and Atlas will turn into an overrun. The shortfall is spread across every cost category, led by Labor.

# Recommendations

The data shows where and when costs ran over, not why. Each recommendation below is a targeted investigation, ranked by urgency.

| Priority | Program | What to investigate | Why now |
|---|---|---|---|
| 1 | Sentinel | Labor hours vs. labor rates, staffing levels, and overtime since August 2024 | The monthly overrun is still growing, about $70K a month by late 2025 |
| 2 | Orion | Materials pricing, supplier contracts, and quantities used vs. plan since June 2024; Subcontracts as a secondary driver | Largest overrun at $1.56M; Materials is 68.6% of it |
| 3 | Atlas | Schedule progress and labor productivity against the plan | Worst CPI (0.954); likely to become an overrun if efficiency does not recover |
| 4 | Portfolio | Add an estimate at completion (EAC) to monthly reporting, using the forecast cost data | Shows where each program is headed, not only where it has been |

For Horizon and Vanguard, keep normal monthly monitoring. Both are over budget, but below the 1.0% Monitor line.

# Limitations and method

### Limitations

- **Synthetic data.** The 864 records (6 programs × 24 months × 6 cost categories) are simulated. The findings show the method, not a real company's results.
- **No root causes in the data.** The dataset has no hours, rates, unit prices, or supplier detail, so the analysis can locate overruns but not explain them.
- **Illustrative thresholds.** The 2.5% / 1.0% budget bands and the 0.975 / 0.990 CPI bands are project choices, not industry standards.
- **"Sustained" is a defined rule.** An overrun is sustained from the first month after the last favorable month.

- **Backward-looking.** Forecast cost was not yet used to project where programs will finish.

**Method**

The analysis moved through three tools, and each one checked the one before it:

1. **Excel:** validated the data, added variance and CPI columns, and found the drivers with PivotTables.
2. **SQL (MySQL):** rebuilt the metrics in a reusable view, ran program, category, and monthly rollups, and applied the Management Status rule.
3. **Power BI:** built a date-table model and DAX measures, and a dashboard that drills from portfolio to program to cost category to month. Every figure matched the SQL results to the cent.

The Power BI model and dashboard were built with AI assistance (Claude, through Microsoft's Power BI authoring tools). The analyst set the direction, made the interpretation calls, and checked every result against the SQL benchmarks.

## Appendix: definitions and thresholds

| Metric | Formula | How to read it |
|---|---|---|
| Budget Variance | Actual Cost − Budget Cost | Positive = over budget (unfavorable) |
| Budget Variance % | Budget Variance ÷ Budget Cost | Makes programs of different sizes comparable |
| Cost Variance | Earned Value − Actual Cost | Negative = less work delivered than paid for |
| CPI | Earned Value ÷ Actual Cost | Below 1.00 = unfavorable cost efficiency |
| Share of net overrun | Program variance ÷ portfolio variance | Negative = offsets the overrun |

| Status | Management Status (Budget Variance %) | CPI Watch (CPI) |
|---|---|---|
| High Attention / Watch | 2.5% or more over budget | Below 0.975 |
| Monitor | 1.0% to 2.5% over budget | 0.975 to 0.990 |

| Status | Management Status (Budget Variance %) | CPI Watch (CPI) |
|---|---|---|
| On Track | Under 1.0% over budget | 0.990 or above |